In [6]:
```
import pandas as pd
```

In [27]:
```
d = {'Name':['Paplu','Taplu','Jhaplu','Oggy','Jack'],
    'Roll':[111,222,333,444,555],
    'Course':['AIML','DA','C++','AIML','Python']}
d
```

Out[27]:
```
{'Name': ['Paplu', 'Taplu', 'Jhaplu', 'Oggy', 'Jack'],
 'Roll': [111, 222, 333, 444, 555],
 'Course': ['AIML', 'DA', 'C++', 'AIML', 'Python']}
```

In [28]:
```
df = pd.DataFrame(d)
df
```

Out[28]:

	Name	Roll	Course
0	Paplu	111	AIML
1	Taplu	222	DA
2	Jhaplu	333	C++
3	Oggy	444	AIML
4	Jack	555	Python

In [29]:
```
#head
df.head()
```

Out[29]:

	Name	Roll	Course
0	Paplu	111	AIML
1	Taplu	222	DA
2	Jhaplu	333	C++
3	Oggy	444	AIML
4	Jack	555	Python

In [30]:
```
#tail
df.tail(3)
```

Out[30]:

	Name	Roll	Course
2	Jhaplu	333	C++
3	Oggy	444	AIML
4	Jack	555	Python

In [31]:
```
#rename
df = df.rename(columns={'Name':'Stu_Name'})
df
```

Out[31]:

	Stu_Name	Roll	Course
0	Paplu	111	AIML
1	Taplu	222	DA
2	Jhaplu	333	C++
3	Oggy	444	AIML
4	Jack	555	Python

In [32]:
```
#set_index
#df = df.set_index("Roll")
#df.set_index("Roll",inplace=True)

df.set_index("Roll",inplace=True)
```

In [33]:
```
df
```

Out[33]:

	Stu_Name	Course
Roll		
111	Paplu	AIML
222	Taplu	DA
333	Jhaplu	C++
444	Oggy	AIML
555	Jack	Python

In [34]:
```
df
```

Out[34]:

	Stu_Name	Course
Roll		
111	Paplu	AIML
222	Taplu	DA
333	Jhaplu	C++
444	Oggy	AIML
555	Jack	Python

In [35]:
```
df.Course[222]
```

Out[35]: 'DA'

In [44]:
```
import pandas as pd
import numpy as np
```

In [45]:
```
d = pd.DataFrame({'Days':[1,2,3,4,5],
                  'Places':['Puri','Manali',np.nan,'BBSR','Goa'],
                  'Visitors':[5890,5000,4580,3800,5555]})
d
```

Out[45]:

	Days	Places	Visitors
0	1	Puri	5890
1	2	Manali	5000
2	3	NaN	4580
3	4	BBSR	3800
4	5	Goa	5555

In [46]:
```
#to check missing values
d.isnull()
```

Out[46]:

	Days	Places	Visitors
0	False	False	False
1	False	False	False
2	False	True	False
3	False	False	False
4	False	False	False

In [47]:
```
d.isnull().sum()
```

Out[47]:
```
Days        0
Places      1
Visitors    0
dtype: int64
```

In [50]:
```
import warnings
warnings.filterwarnings('ignore')
```

In [51]:
```
#fillna
d.Places.fillna('Puri',inplace=True)
```

In [52]:
```
d
```

Out[52]:

	Days	Places	Visitors
0	1	Puri	5890
1	2	Manali	5000
2	3	Puri	4580
3	4	BBSR	3800
4	5	Goa	5555

In [53]:
```
#to check datatypes
d.dtypes
```

Out[53]:
```
Days         int64
Places      object
Visitors     int64
dtype: object
```

In [54]:
```
#info
d.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5 entries, 0 to 4
Data columns (total 3 columns):
 #   Column    Non-Null Count  Dtype 
---  ------    --------------  ----- 
 0   Days      5 non-null      int64 
 1   Places    5 non-null      object
 2   Visitors  5 non-null      int64 
dtypes: int64(2), object(1)
memory usage: 252.0+ bytes
```

In [55]:
```
d.Places.unique()
```

Out[55]:
```
array(['Puri', 'Manali', 'BBSR', 'Goa'], dtype=object)
```

In [56]:
```
d.Visitors.mean()
```

Out[56]:
```
4965.0
```

In [57]:
```
d.Visitors.std()
```

Out[57]:
```
823.1646250902671
```

In [58]:
```
d.Visitors.max()
```

Out[58]:
```
5890
```

In [59]:
```
d.describe()
```

Out[59]:

	Days	Visitors
count	5.000000	5.000000
mean	3.000000	4965.000000
std	1.581139	823.164625
min	1.000000	3800.000000
25%	2.000000	4580.000000
50%	3.000000	5000.000000
75%	4.000000	5555.000000
max	5.000000	5890.000000

In [60]:
```
d.describe(include='all')
```

Out[60]:

	Days	Places	Visitors
count	5.000000	5	5.000000
unique	NaN	4	NaN
top	NaN	Puri	NaN
freq	NaN	2	NaN
mean	3.000000	NaN	4965.000000
std	1.581139	NaN	823.164625
min	1.000000	NaN	3800.000000
25%	2.000000	NaN	4580.000000
50%	3.000000	NaN	5000.000000
75%	4.000000	NaN	5555.000000
max	5.000000	NaN	5890.000000

In [61]:

```
d.describe(include='O')
```

Out[61]:

	Places
count	5
unique	4
top	Puri
freq	2

gTTS - "Google Text To Speech"

In [62]:

```
#to install gtts
!pip3 install gtts
```

```
Collecting gtts
  Using cached gTTS-2.5.4-py3-none-any.whl.metadata (4.1 kB)
Requirement already satisfied: requests<3,>=2.27 in e:\anaconda\lib\site-packages (from gtts)
(2.32.2)
Requirement already satisfied: click<8.2,>=7.1 in e:\anaconda\lib\site-packages (from gtts)
(8.1.7)
Requirement already satisfied: colorama in e:\anaconda\lib\site-packages (from click<8.2,>=7.1
->gtts) (0.4.6)
Requirement already satisfied: charset-normalizer<4,>=2 in e:\anaconda\lib\site-packages (from
requests<3,>=2.27->gtts) (2.0.4)
Requirement already satisfied: idna<4,>=2.5 in e:\anaconda\lib\site-packages (from requests<3,
>=2.27->gtts) (3.7)
Requirement already satisfied: urllib3<3,>=1.21.1 in e:\anaconda\lib\site-packages (from reque
sts<3,>=2.27->gtts) (2.2.2)
Requirement already satisfied: certifi>=2017.4.17 in e:\anaconda\lib\site-packages (from reque
sts<3,>=2.27->gtts) (2025.4.26)
Using cached gTTS-2.5.4-py3-none-any.whl (29 kB)
Installing collected packages: gtts
Successfully installed gtts-2.5.4
```

In [64]:

```
from gtts import gTTS
tts = gTTS("Hello and Welcome Everyone Sipun Kisa Hela Bhalu sanga Sovit Army Boy on Border la
tts.save('hey.mp3')
```

Iris Dataset

In [65]:
```
import numpy as np
import pandas as pd
```

In [108…
```
iris = pd.read_csv(r"C:\Users\CTTC\Downloads\iris (4)\iris.data",header=None)
iris
```

Out[108…

	0	1	2	3	4
0	5.1	3.5	1.4	0.2	Iris-setosa
1	4.9	3.0	1.4	0.2	Iris-setosa
2	4.7	3.2	1.3	0.2	Iris-setosa
3	4.6	3.1	1.5	0.2	Iris-setosa
4	5.0	3.6	1.4	0.2	Iris-setosa
...	...	...	...	...	...
145	6.7	3.0	5.2	2.3	Iris-virginica
146	6.3	2.5	5.0	1.9	Iris-virginica
147	6.5	3.0	5.2	2.0	Iris-virginica
148	6.2	3.4	5.4	2.3	Iris-virginica
149	5.9	3.0	5.1	1.8	Iris-virginica

150 rows × 5 columns

In [109…
```
iris.columns = ['SL','SW','PL','PW','Flower Type']
iris
```

Out[109…

	SL	SW	PL	PW	Flower Type
0	5.1	3.5	1.4	0.2	Iris-setosa
1	4.9	3.0	1.4	0.2	Iris-setosa
2	4.7	3.2	1.3	0.2	Iris-setosa
3	4.6	3.1	1.5	0.2	Iris-setosa
4	5.0	3.6	1.4	0.2	Iris-setosa
...	...	...	...	...	...
145	6.7	3.0	5.2	2.3	Iris-virginica
146	6.3	2.5	5.0	1.9	Iris-virginica
147	6.5	3.0	5.2	2.0	Iris-virginica
148	6.2	3.4	5.4	2.3	Iris-virginica
149	5.9	3.0	5.1	1.8	Iris-virginica

150 rows × 5 columns

In [110…
```
iris.head(10)
```

Out[110…

	SL	SW	PL	PW	Flower Type
0	5.1	3.5	1.4	0.2	Iris-setosa
1	4.9	3.0	1.4	0.2	Iris-setosa
2	4.7	3.2	1.3	0.2	Iris-setosa
3	4.6	3.1	1.5	0.2	Iris-setosa
4	5.0	3.6	1.4	0.2	Iris-setosa
5	5.4	3.9	1.7	0.4	Iris-setosa
6	4.6	3.4	1.4	0.3	Iris-setosa
7	5.0	3.4	1.5	0.2	Iris-setosa
8	4.4	2.9	1.4	0.2	Iris-setosa
9	4.9	3.1	1.5	0.1	Iris-setosa

In [111…

```
iris.head()
```

Out[111…

	SL	SW	PL	PW	Flower Type
0	5.1	3.5	1.4	0.2	Iris-setosa
1	4.9	3.0	1.4	0.2	Iris-setosa
2	4.7	3.2	1.3	0.2	Iris-setosa
3	4.6	3.1	1.5	0.2	Iris-setosa
4	5.0	3.6	1.4	0.2	Iris-setosa

In [112…

```
iris.tail(3)
```

Out[112…

	SL	SW	PL	PW	Flower Type
147	6.5	3.0	5.2	2.0	Iris-virginica
148	6.2	3.4	5.4	2.3	Iris-virginica
149	5.9	3.0	5.1	1.8	Iris-virginica

In [113…

```
#sample
iris.sample()
```

Out[113…

	SL	SW	PL	PW	Flower Type
43	5.0	3.5	1.6	0.6	Iris-setosa

In [114…

```
iris.sample(5)
```

Out[114…

	SL	SW	PL	PW	Flower Type
103	6.3	2.9	5.6	1.8	Iris-virginica
28	5.2	3.4	1.4	0.2	Iris-setosa
74	6.4	2.9	4.3	1.3	Iris-versicolor
113	5.7	2.5	5.0	2.0	Iris-virginica
62	6.0	2.2	4.0	1.0	Iris-versicolor

In [115…

```
iris.iloc[11:21:1, ::]
```

Out[115…

	SL	SW	PL	PW	Flower Type
11	4.8	3.4	1.6	0.2	Iris-setosa
12	4.8	3.0	1.4	0.1	Iris-setosa
13	4.3	3.0	1.1	0.1	Iris-setosa
14	5.8	4.0	1.2	0.2	Iris-setosa
15	5.7	4.4	1.5	0.4	Iris-setosa
16	5.4	3.9	1.3	0.4	Iris-setosa
17	5.1	3.5	1.4	0.3	Iris-setosa
18	5.7	3.8	1.7	0.3	Iris-setosa
19	5.1	3.8	1.5	0.3	Iris-setosa
20	5.4	3.4	1.7	0.2	Iris-setosa

In [116…

```
iris.isnull()
```

Out[116…

	SL	SW	PL	PW	Flower Type
0	False	False	False	False	False
1	False	False	False	False	False
2	False	False	False	False	False
3	False	False	False	False	False
4	False	False	False	False	False
...	...	...	...	...	...
145	False	False	False	False	False
146	False	False	False	False	False
147	False	False	False	False	False
148	False	False	False	False	False
149	False	False	False	False	False

150 rows × 5 columns

In [117…

```
iris.isnull().sum()
```

```
Out[117… SL              0
         SW              0
         PL              0
         PW              0
         Flower Type     0
         dtype: int64
```

```
In [118… iris.dtypes
```

```
Out[118… SL              float64
         SW              float64
         PL              float64
         PW              float64
         Flower Type      object
         dtype: object
```

```
In [119… iris.SL.dtypes
```

```
Out[119… dtype('float64')
```

```
In [120… iris.Flower Type.dtype
```

```
  Cell In[120], line 1
    iris.Flower Type.dtype
                ^
SyntaxError: invalid syntax
```

```
In [121… iris['Flower Type'].dtype
```

```
Out[121… dtype('O')
```

```
In [122… iris['SW'].dtypes
```

```
Out[122… dtype('float64')
```

```
In [123… iris.SL.unique()
```

```
Out[123… array([5.1, 4.9, 4.7, 4.6, 5. , 5.4, 4.4, 4.8, 4.3, 5.8, 5.7, 5.2, 5.5,
                4.5, 5.3, 7. , 6.4, 6.9, 6.5, 6.3, 6.6, 5.9, 6. , 6.1, 5.6, 6.7,
                6.2, 6.8, 7.1, 7.6, 7.3, 7.2, 7.7, 7.4, 7.9])
```

```
In [124… iris['Flower Type'].unique()
```

```
Out[124… array(['Iris-setosa', 'Iris-versicolor', 'Iris-virginica'], dtype=object)
```

```
In [125… iris.columns
```

```
Out[125… Index(['SL', 'SW', 'PL', 'PW', 'Flower Type'], dtype='object')
```

```
In [126… for i in iris.columns:
             print(i,':','\n',iris[i].unique(),'\n')
```

```
SL :
 [5.1 4.9 4.7 4.6 5.  5.4 4.4 4.8 4.3 5.8 5.7 5.2 5.5 4.5 5.3 7.  6.4 6.9
 6.5 6.3 6.6 5.9 6.  6.1 5.6 6.7 6.2 6.8 7.1 7.6 7.3 7.2 7.7 7.4 7.9]

SW :
 [3.5 3.  3.2 3.1 3.6 3.9 3.4 2.9 3.7 4.  4.4 3.8 3.3 4.1 4.2 2.3 2.8 2.4
 2.7 2.  2.2 2.5 2.6]

PL :
 [1.4 1.3 1.5 1.7 1.6 1.1 1.2 1.  1.9 4.7 4.5 4.9 4.  4.6 3.3 3.9 3.5 4.2
 3.6 4.4 4.1 4.8 4.3 5.  3.8 3.7 5.1 3.  6.  5.9 5.6 5.8 6.6 6.3 6.1 5.3
 5.5 6.7 6.9 5.7 6.4 5.4 5.2]

PW :
 [0.2 0.4 0.3 0.1 0.5 0.6 1.4 1.5 1.3 1.6 1.  1.1 1.8 1.2 1.7 2.5 1.9 2.1
 2.2 2.  2.4 2.3]

Flower Type :
 ['Iris-setosa' 'Iris-versicolor' 'Iris-virginica']
```

In [127…

```
iris.describe()
```

Out[127…

	SL	SW	PL	PW
count	150.000000	150.000000	150.000000	150.000000
mean	5.843333	3.054000	3.758667	1.198667
std	0.828066	0.433594	1.764420	0.763161
min	4.300000	2.000000	1.000000	0.100000
25%	5.100000	2.800000	1.600000	0.300000
50%	5.800000	3.000000	4.350000	1.300000
75%	6.400000	3.300000	5.100000	1.800000
max	7.900000	4.400000	6.900000	2.500000

In [128…

```
iris.describe(include='all')
```

Out[128…

	SL	SW	PL	PW	Flower Type
count	150.000000	150.000000	150.000000	150.000000	150
unique	NaN	NaN	NaN	NaN	3
top	NaN	NaN	NaN	NaN	Iris-setosa
freq	NaN	NaN	NaN	NaN	50
mean	5.843333	3.054000	3.758667	1.198667	NaN
std	0.828066	0.433594	1.764420	0.763161	NaN
min	4.300000	2.000000	1.000000	0.100000	NaN
25%	5.100000	2.800000	1.600000	0.300000	NaN
50%	5.800000	3.000000	4.350000	1.300000	NaN
75%	6.400000	3.300000	5.100000	1.800000	NaN
max	7.900000	4.400000	6.900000	2.500000	NaN

In [129…
```python
iris.describe(include='O')
```

Out[129…

	Flower Type
count	150
unique	3
top	Iris-setosa
freq	50

In [130…
```python
iris['Flower Type'].value_counts()
```

Out[130…
```
Flower Type
Iris-setosa        50
Iris-versicolor    50
Iris-virginica     50
Name: count, dtype: int64
```

In [131…
```python
iris.SL.value_counts()
```

Out[131…
```
SL
5.0    10
5.1     9
6.3     9
5.7     8
6.7     8
5.8     7
5.5     7
6.4     7
4.9     6
5.4     6
6.1     6
6.0     6
5.6     6
4.8     5
6.5     5
6.2     4
7.7     4
6.9     4
4.6     4
5.2     4
5.9     3
4.4     3
7.2     3
6.8     3
6.6     2
4.7     2
7.6     1
7.4     1
7.3     1
7.0     1
7.1     1
5.3     1
4.3     1
4.5     1
7.9     1
Name: count, dtype: int64
```

In [133…
```python
iris.drop('Flower Type',axis=1,inplace=True)
iris
```

Out[133…

	SL	SW	PL	PW
0	5.1	3.5	1.4	0.2
1	4.9	3.0	1.4	0.2
2	4.7	3.2	1.3	0.2
3	4.6	3.1	1.5	0.2
4	5.0	3.6	1.4	0.2
...	...	...	...	...
145	6.7	3.0	5.2	2.3
146	6.3	2.5	5.0	1.9
147	6.5	3.0	5.2	2.0
148	6.2	3.4	5.4	2.3
149	5.9	3.0	5.1	1.8

150 rows × 4 columns

AutoMPG

In [1]:

```
#importing packages
import numpy as np
import pandas as pd
```

In [6]:

```
#Reading the Dataset
#delim_whitespace = True or sep='\s+'
auto = pd.read_csv(r"C:\Users\CTTC\Downloads\auto+mpg\auto-mpg.data",header=None,sep='\s+')
auto
```

```
<>:3: SyntaxWarning: invalid escape sequence '\s'
<>:3: SyntaxWarning: invalid escape sequence '\s'
C:\Users\CTTC\AppData\Local\Temp\ipykernel_4344\927710187.py:3: SyntaxWarning: invalid escape
sequence '\s'
  auto = pd.read_csv(r"C:\Users\CTTC\Downloads\auto+mpg\auto-mpg.data",header=None,sep='\s+')
```

Out[6]:

	0	1	2	3	4	5	6	7	8
0	18.0	8	307.0	130.0	3504.0	12.0	70	1	chevrolet chevelle malibu
1	15.0	8	350.0	165.0	3693.0	11.5	70	1	buick skylark 320
2	18.0	8	318.0	150.0	3436.0	11.0	70	1	plymouth satellite
3	16.0	8	304.0	150.0	3433.0	12.0	70	1	amc rebel sst
4	17.0	8	302.0	140.0	3449.0	10.5	70	1	ford torino
...	...	...	...	...	...	...	...	...	...
393	27.0	4	140.0	86.00	2790.0	15.6	82	1	ford mustang gl
394	44.0	4	97.0	52.00	2130.0	24.6	82	2	vw pickup
395	32.0	4	135.0	84.00	2295.0	11.6	82	1	dodge rampage
396	28.0	4	120.0	79.00	2625.0	18.6	82	1	ford ranger
397	31.0	4	119.0	82.00	2720.0	19.4	82	1	chevy s-10

398 rows × 9 columns

In [8]:

```
#Renaming the columns (if required)
auto.columns = ['mpg','cylinders','displacement',
               'horsepower','weight','acceleration',
               'model year','origin','car name']
auto.head()
```

Out[8]:

	mpg	cylinders	displacement	horsepower	weight	acceleration	model year	origin	car name
0	18.0	8	307.0	130.0	3504.0	12.0	70	1	chevrolet chevelle malibu
1	15.0	8	350.0	165.0	3693.0	11.5	70	1	buick skylark 320
2	18.0	8	318.0	150.0	3436.0	11.0	70	1	plymouth satellite
3	16.0	8	304.0	150.0	3433.0	12.0	70	1	amc rebel sst
4	17.0	8	302.0	140.0	3449.0	10.5	70	1	ford torino

In [9]:

```
#to check missing values
auto.isnull().sum()
```

Out[9]:

```
mpg             0
cylinders       0
displacement    0
horsepower      0
weight          0
acceleration    0
model year      0
origin          0
car name        0
dtype: int64
```

In [10]:

```
#to check datatypes
auto.dtypes
```

Out[10]:
```
mpg             float64
cylinders         int64
displacement    float64
horsepower       object
weight          float64
acceleration    float64
model year        int64
origin            int64
car name         object
dtype: object
```

In [11]:
```
#to check unique values
for i in auto.columns:
    print(i,':','\n',auto[i].unique(),'\n')
```

```
mpg :
 [18.  15.  16.  17.  14.  24.  22.  21.  27.  26.  25.  10.  11.   9.
 28.  19.  12.  13.  23.  30.  31.  35.  20.  29.  32.  33.  17.5 15.5
 14.5 22.5 24.5 18.5 29.5 26.5 16.5 31.5 36.  25.5 33.5 20.5 30.5 21.5
 43.1 36.1 32.8 39.4 19.9 19.4 20.2 19.2 25.1 20.6 20.8 18.6 18.1 17.7
 27.5 27.2 30.9 21.1 23.2 23.8 23.9 20.3 21.6 16.2 19.8 22.3 17.6 18.2
 16.9 31.9 34.1 35.7 27.4 25.4 34.2 34.5 31.8 37.3 28.4 28.8 26.8 41.5
 38.1 32.1 37.2 26.4 24.3 19.1 34.3 29.8 31.3 37.  32.2 46.6 27.9 40.8
 44.3 43.4 36.4 44.6 40.9 33.8 32.7 23.7 23.6 32.4 26.6 25.8 23.5 39.1
 39.  35.1 32.3 37.7 34.7 34.4 29.9 33.7 32.9 31.6 28.1 30.7 24.2 22.4
 34.  38.  44. ]

cylinders :
 [8 4 6 3 5]

displacement :
 [307.  350.  318.  304.  302.  429.  454.  440.  455.  390.  383.  340.
 400.  113.  198.  199.  200.   97.  110.  107.  104.  121.  360.  140.
  98.  232.  225.  250.  351.  258.  122.  116.   79.   88.   71.   72.
  91.   97.5  70.  120.   96.  108.  155.   68.  114.  156.   76.   83.
  90.  231.  262.  134.  119.  171.  115.  101.  305.   85.  130.  168.
 111.  260.  151.  146.   80.   78.  105.  131.  163.   89.  267.   86.
 183.  141.  173.  135.   81.  100.  145.  112.  181.  144. ]

horsepower :
 ['130.0' '165.0' '150.0' '140.0' '198.0' '220.0' '215.0' '225.0' '190.0'
 '170.0' '160.0' '95.00' '97.00' '85.00' '88.00' '46.00' '87.00' '90.00'
 '113.0' '200.0' '210.0' '193.0' '?' '100.0' '105.0' '175.0' '153.0'
 '180.0' '110.0' '72.00' '86.00' '70.00' '76.00' '65.00' '69.00' '60.00'
 '80.00' '54.00' '208.0' '155.0' '112.0' '92.00' '145.0' '137.0' '158.0'
 '167.0' '94.00' '107.0' '230.0' '49.00' '75.00' '91.00' '122.0' '67.00'
 '83.00' '78.00' '52.00' '61.00' '93.00' '148.0' '129.0' '96.00' '71.00'
 '98.00' '115.0' '53.00' '81.00' '79.00' '120.0' '152.0' '102.0' '108.0'
 '68.00' '58.00' '149.0' '89.00' '63.00' '48.00' '66.00' '139.0' '103.0'
 '125.0' '133.0' '138.0' '135.0' '142.0' '77.00' '62.00' '132.0' '84.00'
 '64.00' '74.00' '116.0' '82.00']

weight :
 [3504. 3693. 3436. 3433. 3449. 4341. 4354. 4312. 4425. 3850. 3563. 3609.
 3761. 3086. 2372. 2833. 2774. 2587. 2130. 1835. 2672. 2430. 2375. 2234.
 2648. 4615. 4376. 4382. 4732. 2264. 2228. 2046. 2634. 3439. 3329. 3302.
 3288. 4209. 4464. 4154. 4096. 4955. 4746. 5140. 2962. 2408. 3282. 3139.
 2220. 2123. 2074. 2065. 1773. 1613. 1834. 1955. 2278. 2126. 2254. 2226.
 4274. 4385. 4135. 4129. 3672. 4633. 4502. 4456. 4422. 2330. 3892. 4098.
 4294. 4077. 2933. 2511. 2979. 2189. 2395. 2288. 2506. 2164. 2100. 4100.
 3988. 4042. 3777. 4952. 4363. 4237. 4735. 4951. 3821. 3121. 3278. 2945.
 3021. 2904. 1950. 4997. 4906. 4654. 4499. 2789. 2279. 2401. 2379. 2124.
 2310. 2472. 2265. 4082. 4278. 1867. 2158. 2582. 2868. 3399. 2660. 2807.
 3664. 3102. 2875. 2901. 3336. 2451. 1836. 2542. 3781. 3632. 3613. 4141.
 4699. 4457. 4638. 4257. 2219. 1963. 2300. 1649. 2003. 2125. 2108. 2246.
 2489. 2391. 2000. 3264. 3459. 3432. 3158. 4668. 4440. 4498. 4657. 3907.
 3897. 3730. 3785. 3039. 3221. 3169. 2171. 2639. 2914. 2592. 2702. 2223.
 2545. 2984. 1937. 3211. 2694. 2957. 2671. 1795. 2464. 2572. 2255. 2202.
 4215. 4190. 3962. 3233. 3353. 3012. 3085. 2035. 3651. 3574. 3645. 3193.
 1825. 1990. 2155. 2565. 3150. 3940. 3270. 2930. 3820. 4380. 4055. 3870.
 3755. 2045. 1945. 3880. 4060. 4140. 4295. 3520. 3425. 3630. 3525. 4220.
 4165. 4325. 4335. 1940. 2740. 2755. 2051. 2075. 1985. 2190. 2815. 2600.
 2720. 1800. 2070. 3365. 3735. 3570. 3535. 3155. 2965. 3430. 3210. 3380.
 3070. 3620. 3410. 3445. 3205. 4080. 2560. 2230. 2515. 2745. 2855. 2405.
 2830. 3140. 2795. 2135. 3245. 2990. 2890. 3265. 3360. 3840. 3725. 3955.
 3830. 4360. 4054. 3605. 1925. 1975. 1915. 2670. 3530. 3900. 3190. 3420.
 2200. 2150. 2020. 2595. 2700. 2556. 2144. 1968. 2120. 2019. 2678. 2870.
 3003. 3381. 2188. 2711. 2434. 2110. 2800. 2085. 2335. 2950. 3250. 1850.
 2145. 1845. 2910. 2420. 2500. 2905. 2290. 2490. 2635. 2620. 2725. 2385.
 1755. 1875. 1760. 2050. 2215. 2380. 2320. 2210. 2350. 2615. 3230. 3160.
 2900. 3415. 3060. 3465. 2605. 2640. 2575. 2525. 2735. 2865. 3035. 1980.
```

```
 2025. 1970. 2160. 2205. 2245. 1965. 1995. 3015. 2585. 2835. 2665. 2370.
 2790. 2295. 2625.]

acceleration :
 [12.  11.5 11.  10.5 10.   9.   8.5  8.   9.5 15.  15.5 16.  14.5 20.5
 17.5 12.5 14.  13.5 18.5 19.  13.  19.5 18.  17.  23.5 16.5 21.  16.9
 14.9 17.7 15.3 13.9 12.8 15.4 17.6 22.2 22.1 14.2 17.4 16.2 17.8 12.2
 16.4 13.6 15.7 13.2 21.9 16.7 12.1 14.8 18.6 16.8 13.7 11.1 11.4 18.2
 15.8 15.9 14.1 21.5 14.4 19.4 19.2 17.2 18.7 15.1 13.4 11.2 14.7 16.6
 17.3 15.2 14.3 20.1 24.8 11.3 12.9 18.8 18.1 17.9 21.7 23.7 19.9 21.8
 13.8 12.6 16.1 20.7 18.3 20.4 19.6 17.1 15.6 24.6 11.6]

model year :
 [70 71 72 73 74 75 76 77 78 79 80 81 82]

origin :
 [1 3 2]

car name :
 ['chevrolet chevelle malibu' 'buick skylark 320' 'plymouth satellite'
 'amc rebel sst' 'ford torino' 'ford galaxie 500' 'chevrolet impala'
 'plymouth fury iii' 'pontiac catalina' 'amc ambassador dpl'
 'dodge challenger se' "plymouth 'cuda 340" 'chevrolet monte carlo'
 'buick estate wagon (sw)' 'toyota corona mark ii' 'plymouth duster'
 'amc hornet' 'ford maverick' 'datsun pl510'
 'volkswagen 1131 deluxe sedan' 'peugeot 504' 'audi 100 ls' 'saab 99e'
 'bmw 2002' 'amc gremlin' 'ford f250' 'chevy c20' 'dodge d200' 'hi 1200d'
 'chevrolet vega 2300' 'toyota corona' 'ford pinto'
 'plymouth satellite custom' 'ford torino 500' 'amc matador'
 'pontiac catalina brougham' 'dodge monaco (sw)'
 'ford country squire (sw)' 'pontiac safari (sw)'
 'amc hornet sportabout (sw)' 'chevrolet vega (sw)' 'pontiac firebird'
 'ford mustang' 'mercury capri 2000' 'opel 1900' 'peugeot 304' 'fiat 124b'
 'toyota corolla 1200' 'datsun 1200' 'volkswagen model 111'
 'plymouth cricket' 'toyota corona hardtop' 'dodge colt hardtop'
 'volkswagen type 3' 'chevrolet vega' 'ford pinto runabout'
 'amc ambassador sst' 'mercury marquis' 'buick lesabre custom'
 'oldsmobile delta 88 royale' 'chrysler newport royal' 'mazda rx2 coupe'
 'amc matador (sw)' 'chevrolet chevelle concours (sw)'
 'ford gran torino (sw)' 'plymouth satellite custom (sw)'
 'volvo 145e (sw)' 'volkswagen 411 (sw)' 'peugeot 504 (sw)'
 'renault 12 (sw)' 'ford pinto (sw)' 'datsun 510 (sw)'
 'toyouta corona mark ii (sw)' 'dodge colt (sw)'
 'toyota corolla 1600 (sw)' 'buick century 350' 'chevrolet malibu'
 'ford gran torino' 'dodge coronet custom' 'mercury marquis brougham'
 'chevrolet caprice classic' 'ford ltd' 'plymouth fury gran sedan'
 'chrysler new yorker brougham' 'buick electra 225 custom'
 'amc ambassador brougham' 'plymouth valiant' 'chevrolet nova custom'
 'volkswagen super beetle' 'ford country' 'plymouth custom suburb'
 'oldsmobile vista cruiser' 'toyota carina' 'datsun 610' 'maxda rx3'
 'mercury capri v6' 'fiat 124 sport coupe' 'chevrolet monte carlo s'
 'pontiac grand prix' 'fiat 128' 'opel manta' 'audi 100ls' 'volvo 144ea'
 'dodge dart custom' 'saab 99le' 'toyota mark ii' 'oldsmobile omega'
 'chevrolet nova' 'datsun b210' 'chevrolet chevelle malibu classic'
 'plymouth satellite sebring' 'buick century luxus (sw)'
 'dodge coronet custom (sw)' 'audi fox' 'volkswagen dasher' 'datsun 710'
 'dodge colt' 'fiat 124 tc' 'honda civic' 'subaru' 'fiat x1.9'
 'plymouth valiant custom' 'mercury monarch' 'chevrolet bel air'
 'plymouth grand fury' 'buick century' 'chevroelt chevelle malibu'
 'plymouth fury' 'buick skyhawk' 'chevrolet monza 2+2' 'ford mustang ii'
 'toyota corolla' 'pontiac astro' 'volkswagen rabbit' 'amc pacer'
 'volvo 244dl' 'honda civic cvcc' 'fiat 131' 'capri ii' 'renault 12tl'
 'dodge coronet brougham' 'chevrolet chevette' 'chevrolet woody'
 'vw rabbit' 'dodge aspen se' 'ford granada ghia' 'pontiac ventura sj'
 'amc pacer d/l' 'datsun b-210' 'volvo 245' 'plymouth volare premier v8'
 'mercedes-benz 280s' 'cadillac seville' 'chevy c10' 'ford f108'
```

```
 'dodge d100' 'honda accord cvcc' 'buick opel isuzu deluxe'
 'renault 5 gtl' 'plymouth arrow gs' 'datsun f-10 hatchback'
 'oldsmobile cutlass supreme' 'dodge monaco brougham'
 'mercury cougar brougham' 'chevrolet concours' 'buick skylark'
 'plymouth volare custom' 'ford granada' 'pontiac grand prix lj'
 'chevrolet monte carlo landau' 'chrysler cordoba' 'ford thunderbird'
 'volkswagen rabbit custom' 'pontiac sunbird coupe'
 'toyota corolla liftback' 'ford mustang ii 2+2' 'dodge colt m/m'
 'subaru dl' 'datsun 810' 'bmw 320i' 'mazda rx-4'
 'volkswagen rabbit custom diesel' 'ford fiesta' 'mazda glc deluxe'
 'datsun b210 gx' 'oldsmobile cutlass salon brougham' 'dodge diplomat'
 'mercury monarch ghia' 'pontiac phoenix lj' 'ford fairmont (auto)'
 'ford fairmont (man)' 'plymouth volare' 'amc concord'
 'buick century special' 'mercury zephyr' 'dodge aspen' 'amc concord d/l'
 'buick regal sport coupe (turbo)' 'ford futura' 'dodge magnum xe'
 'datsun 510' 'dodge omni' 'toyota celica gt liftback' 'plymouth sapporo'
 'oldsmobile starfire sx' 'datsun 200-sx' 'audi 5000' 'volvo 264gl'
 'saab 99gle' 'peugeot 604sl' 'volkswagen scirocco' 'honda accord lx'
 'pontiac lemans v6' 'mercury zephyr 6' 'ford fairmont 4'
 'amc concord dl 6' 'dodge aspen 6' 'ford ltd landau'
 'mercury grand marquis' 'dodge st. regis' 'chevrolet malibu classic (sw)'
 'chrysler lebaron town @ country (sw)' 'vw rabbit custom'
 'maxda glc deluxe' 'dodge colt hatchback custom' 'amc spirit dl'
 'mercedes benz 300d' 'cadillac eldorado' 'plymouth horizon'
 'plymouth horizon tc3' 'datsun 210' 'fiat strada custom'
 'buick skylark limited' 'chevrolet citation' 'oldsmobile omega brougham'
 'pontiac phoenix' 'toyota corolla tercel' 'datsun 310' 'ford fairmont'
 'audi 4000' 'toyota corona liftback' 'mazda 626' 'datsun 510 hatchback'
 'mazda glc' 'vw rabbit c (diesel)' 'vw dasher (diesel)'
 'audi 5000s (diesel)' 'mercedes-benz 240d' 'honda civic 1500 gl'
 'renault lecar deluxe' 'vokswagen rabbit' 'datsun 280-zx' 'mazda rx-7 gs'
 'triumph tr7 coupe' 'ford mustang cobra' 'honda accord'
 'plymouth reliant' 'dodge aries wagon (sw)' 'toyota starlet'
 'plymouth champ' 'honda civic 1300' 'datsun 210 mpg' 'toyota tercel'
 'mazda glc 4' 'plymouth horizon 4' 'ford escort 4w' 'ford escort 2h'
 'volkswagen jetta' 'renault 18i' 'honda prelude' 'datsun 200sx'
 'peugeot 505s turbo diesel' 'volvo diesel' 'toyota cressida'
 'datsun 810 maxima' 'oldsmobile cutlass ls' 'ford granada gl'
 'chrysler lebaron salon' 'chevrolet cavalier' 'chevrolet cavalier wagon'
 'chevrolet cavalier 2-door' 'pontiac j2000 se hatchback' 'dodge aries se'
 'ford fairmont futura' 'amc concord dl' 'volkswagen rabbit l'
 'mazda glc custom l' 'mazda glc custom' 'plymouth horizon miser'
 'mercury lynx l' 'nissan stanza xe' 'honda civic (auto)' 'datsun 310 gx'
 'buick century limited' 'oldsmobile cutlass ciera (diesel)'
 'chrysler lebaron medallion' 'ford granada l' 'toyota celica gt'
 'dodge charger 2.2' 'chevrolet camaro' 'ford mustang gl' 'vw pickup'
 'dodge rampage' 'ford ranger' 'chevy s-10']
```

In [13]:

```
for i in auto.columns:
    print(i,':',sum(auto[i]=='?'))
```

```
mpg : 0
cylinders : 0
displacement : 0
horsepower : 6
weight : 0
acceleration : 0
model year : 0
origin : 0
car name : 0
```

In [14]:

```
#describe
d = auto.describe(include='all')
d
```

Out[14]:

	mpg	cylinders	displacement	horsepower	weight	acceleration	model year	
count	398.000000	398.000000	398.000000	398	398.000000	398.000000	398.000000	398
unique	NaN	NaN	NaN	94	NaN	NaN	NaN	
top	NaN	NaN	NaN	150.0	NaN	NaN	NaN	
freq	NaN	NaN	NaN	22	NaN	NaN	NaN	
mean	23.514573	5.454774	193.425879	NaN	2970.424623	15.568090	76.010050	1
std	7.815984	1.701004	104.269838	NaN	846.841774	2.757689	3.697627	0
min	9.000000	3.000000	68.000000	NaN	1613.000000	8.000000	70.000000	1
25%	17.500000	4.000000	104.250000	NaN	2223.750000	13.825000	73.000000	1
50%	23.000000	4.000000	148.500000	NaN	2803.500000	15.500000	76.000000	1
75%	29.000000	8.000000	262.000000	NaN	3608.000000	17.175000	79.000000	2
max	46.600000	8.000000	455.000000	NaN	5140.000000	24.800000	82.000000	3

In [15]:

```
auto.horsepower.replace('?',d['horsepower'][2],inplace=True)
```

```
C:\Users\CTTC\AppData\Local\Temp\ipykernel_4344\227787081.py:1: FutureWarning: Series.__getite
m__ treating keys as positions is deprecated. In a future version, integer keys will always be
treated as labels (consistent with DataFrame behavior). To access a value by position, use `se
r.iloc[pos]`
  auto.horsepower.replace('?',d['horsepower'][2],inplace=True)
```

In [16]:

```
for i in auto.columns:
    print(i,':',sum(auto[i]=='?'))
```

```
mpg : 0
cylinders : 0
displacement : 0
horsepower : 0
weight : 0
acceleration : 0
model year : 0
origin : 0
car name : 0
```

In [17]:

```
auto.dtypes
```

Out[17]:

```
mpg             float64
cylinders         int64
displacement    float64
horsepower       object
weight          float64
acceleration    float64
model year        int64
origin            int64
car name         object
dtype: object
```

In [18]:

```
auto.horsepower = auto.horsepower.astype(float)
auto.dtypes
```

```
Out[18]: mpg               float64
         cylinders           int64
         displacement      float64
         horsepower        float64
         weight            float64
         acceleration      float64
         model year          int64
         origin              int64
         car name           object
         dtype: object
```

Matplotlib

```
Data Visualization Technique

**Markers**

=============   ===============================
character       description
=============   ===============================
``'.'``         point marker
``','``         pixel marker
``'o'``         circle marker
``'v'``         triangle_down marker
``'^'``         triangle_up marker
``'<'``         triangle_left marker
``'>'``         triangle_right marker
``'1'``         tri_down marker
``'2'``         tri_up marker
``'3'``         tri_left marker
``'4'``         tri_right marker
``'8'``         octagon marker
``'s'``         square marker
``'p'``         pentagon marker
``'P'``         plus (filled) marker
``'*'``         star marker
``'h'``         hexagon1 marker
``'H'``         hexagon2 marker
``'+'``         plus marker
``'x'``         x marker
``'X'``         x (filled) marker
``'D'``         diamond marker
``'d'``         thin_diamond marker
``'|'``         vline marker
``'_'``         hline marker
=============   ===============================

**Line Styles**

=============    ===============================
character        description
=============    ===============================
``'-'``          solid line style
``'--'``         dashed line style
``'-.'``         dash-dot line style
``':'``          dotted line style
=============    ===============================

Example format strings::
```

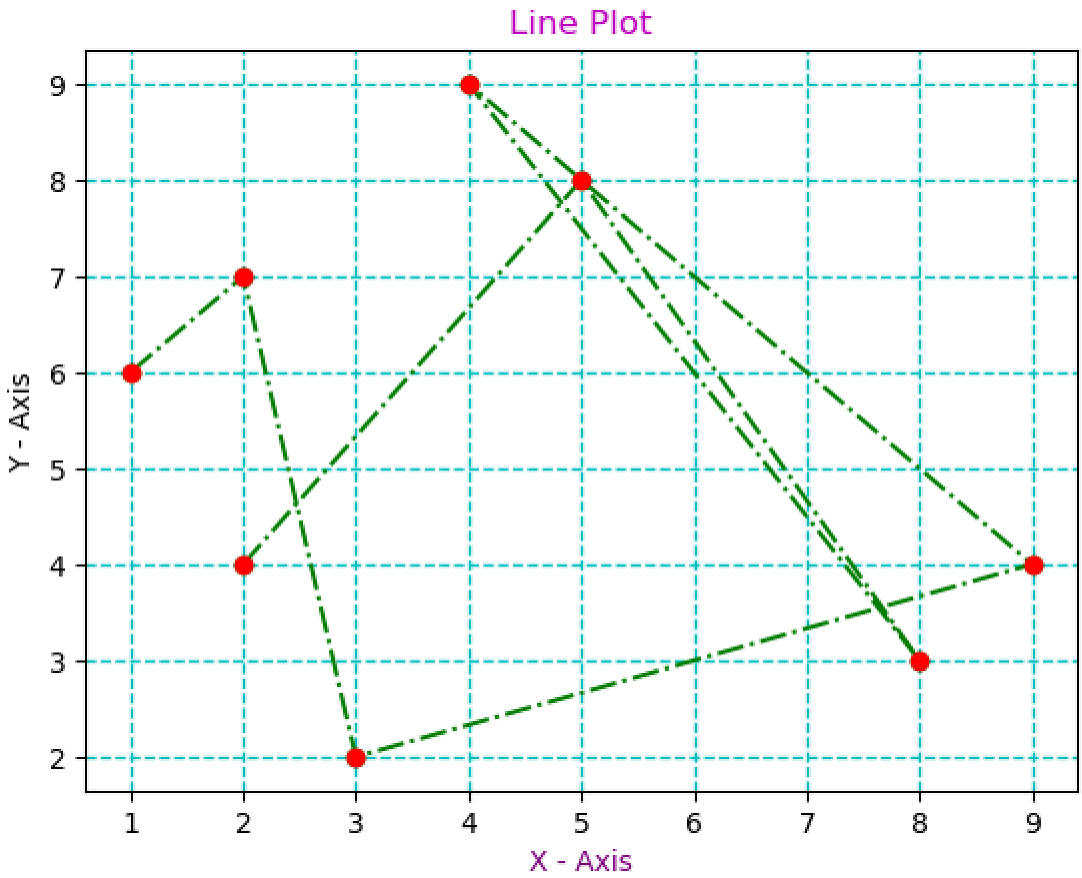
```
        'b'    # blue markers with default shape
        'or'   # red circles
        '-g'   # green solid line
        '--'   # dashed line with default color
        '^k:'  # black triangle_up markers connected by a dotted line

    **Colors**

    The supported color abbreviations are the single letter codes

    =============    ===============================
    character        color
    =============    ===============================
    ``'b'``          blue
    ``'g'``          green
    ``'r'``          red
    ``'c'``          cyan
    ``'m'``          magenta
    ``'y'``          yellow
    ``'k'``          black
    ``'w'``          white
    =============    ===============================
```

In [19]:

```
import matplotlib.pyplot as plt
```

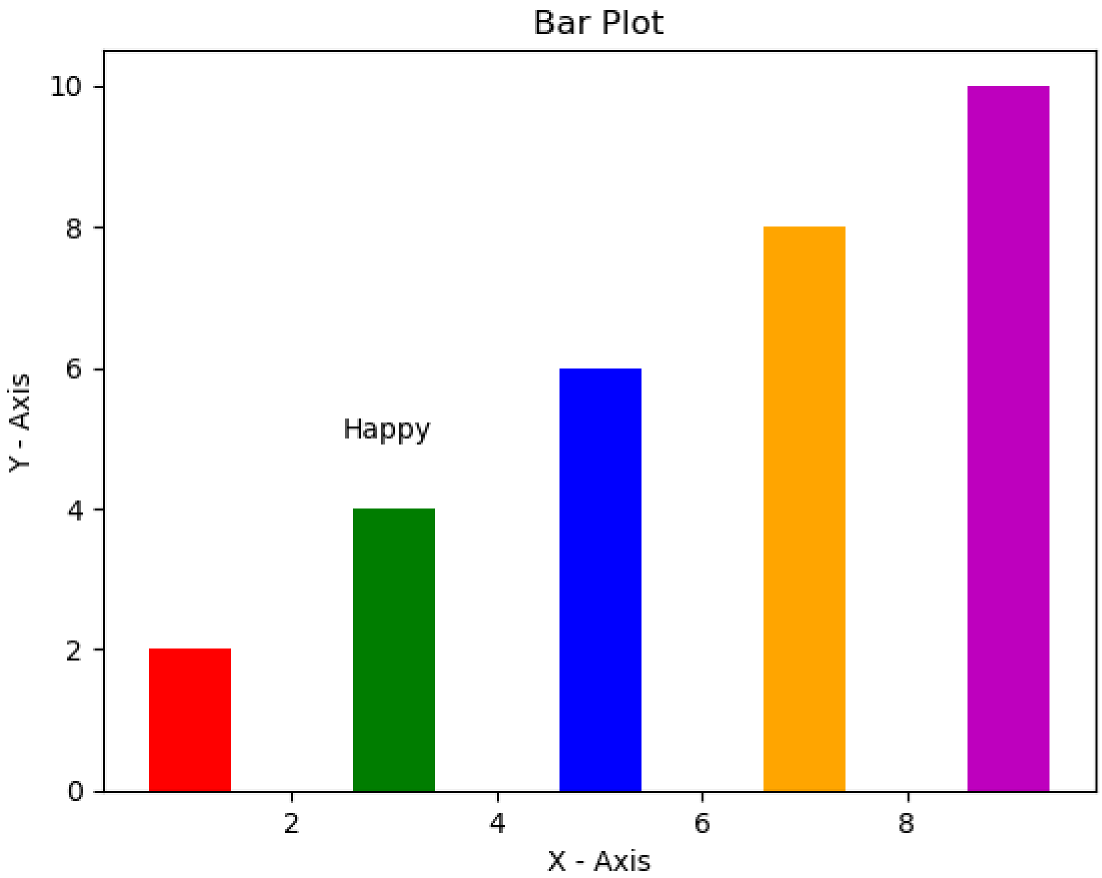
In [47]:

```
#Line Plot
x = [2,5,8,4,9,3,2,1]
y = [4,8,3,9,4,2,7,6]

plt.plot(x,y,'og-.')
plt.plot(x,y,'or')
plt.title("Line Plot",color='m')
plt.xlabel('X - Axis',color='purple')
plt.ylabel('Y - Axis')
plt.grid(color='c',linestyle='--', linewidth=1)
plt.show()
```

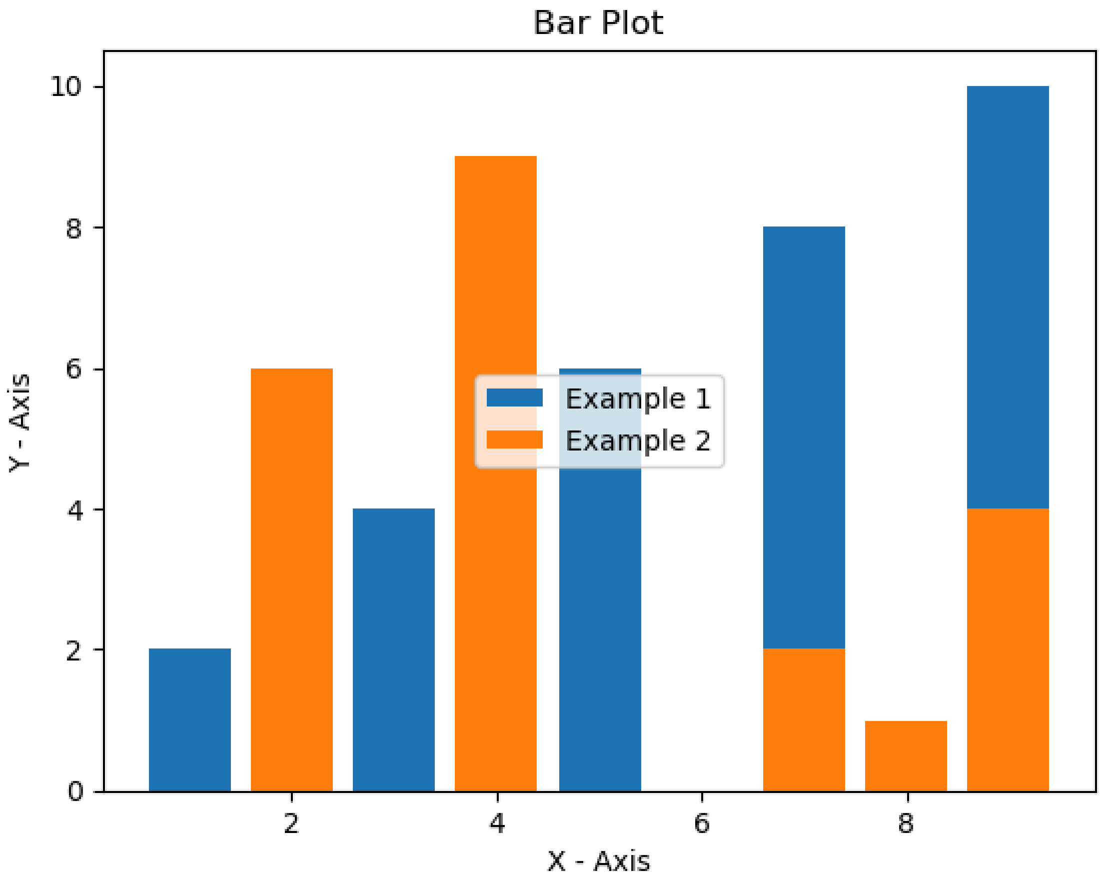


In [54]:

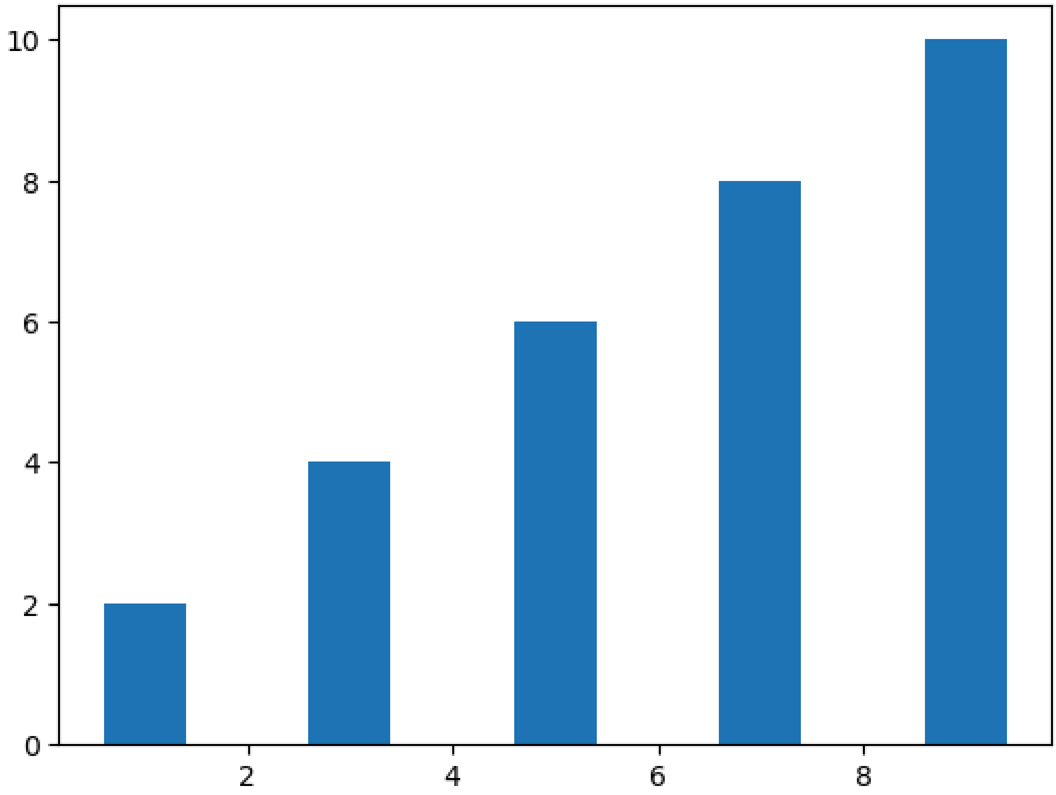
```
#Bar Plot
x = [1,3,5,7,9]
y = [2,4,6,8,10]
cols = ['r','g','b','orange','m']

plt.bar(x,y,color=cols)
plt.title("Bar Plot")
plt.xlabel('X - Axis')
plt.ylabel('Y - Axis')
plt.annotate("Happy",xy=(2.5,5))
plt.show()
```

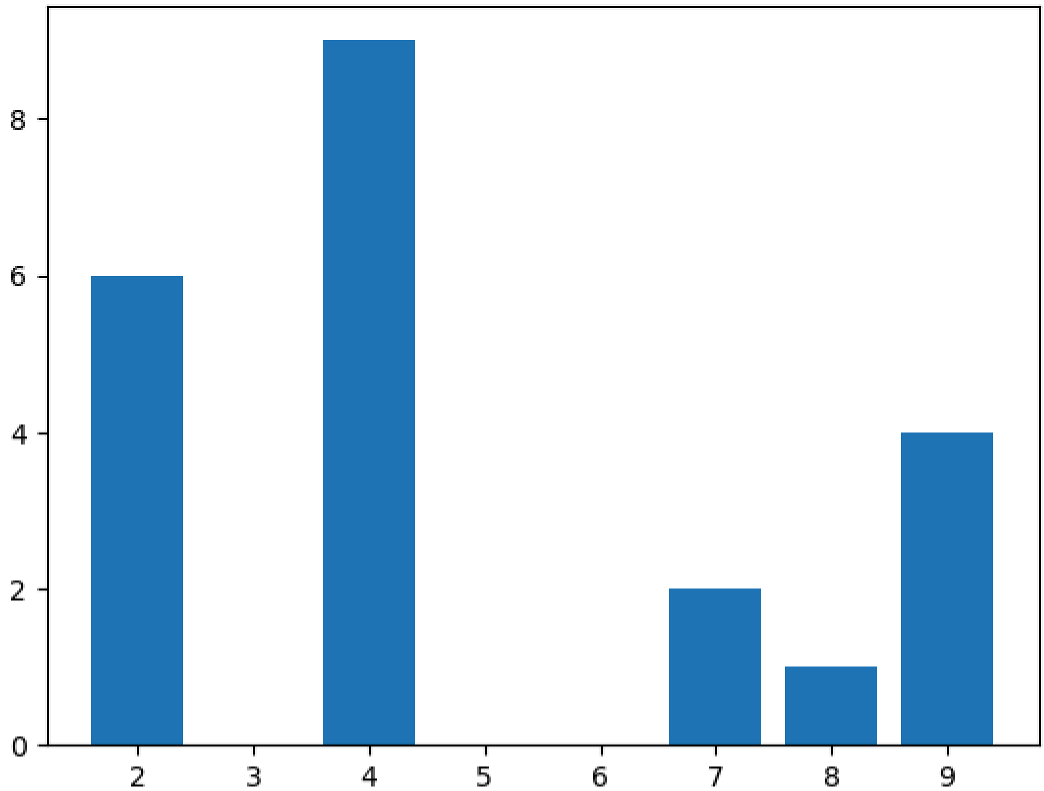


In [59]:
```
#Multiplt Bar Plot
x = [1,3,5,7,9]
y = [2,4,6,8,10]
plt.bar(x,y,label='Example 1')

x1 = [4,7,2,9,8]
y1 = [9,2,6,4,1]
plt.bar(x1,y1,label='Example 2')

plt.title("Bar Plot")
plt.legend(loc='center')
plt.xlabel('X - Axis')
plt.ylabel('Y - Axis')
plt.show()
```

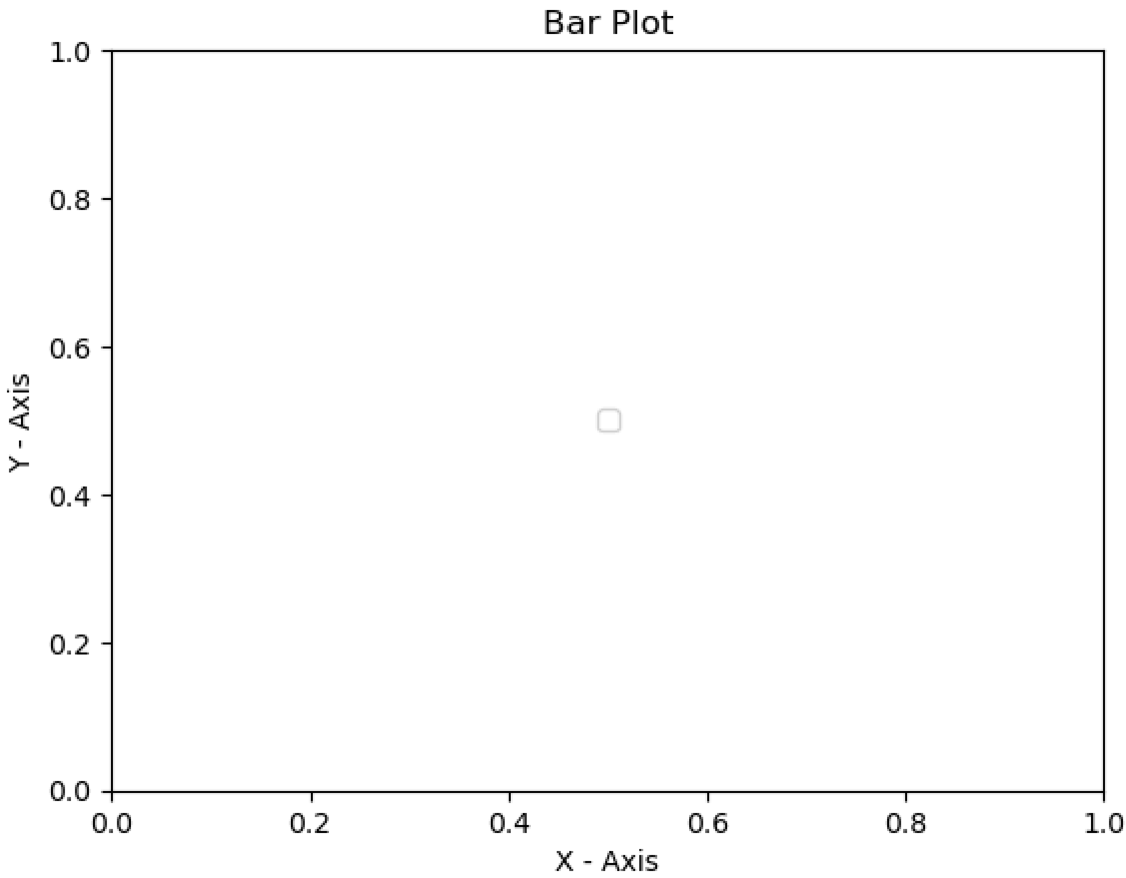


In [60]:
```
#Multiplt Bar Plot
x = [1,3,5,7,9]
y = [2,4,6,8,10]
plt.bar(x,y,label='Example 1')
plt.show()

x1 = [4,7,2,9,8]
y1 = [9,2,6,4,1]
plt.bar(x1,y1,label='Example 2')
plt.show()

plt.title("Bar Plot")
plt.legend(loc='center')
plt.xlabel('X - Axis')
plt.ylabel('Y - Axis')
plt.show()
```

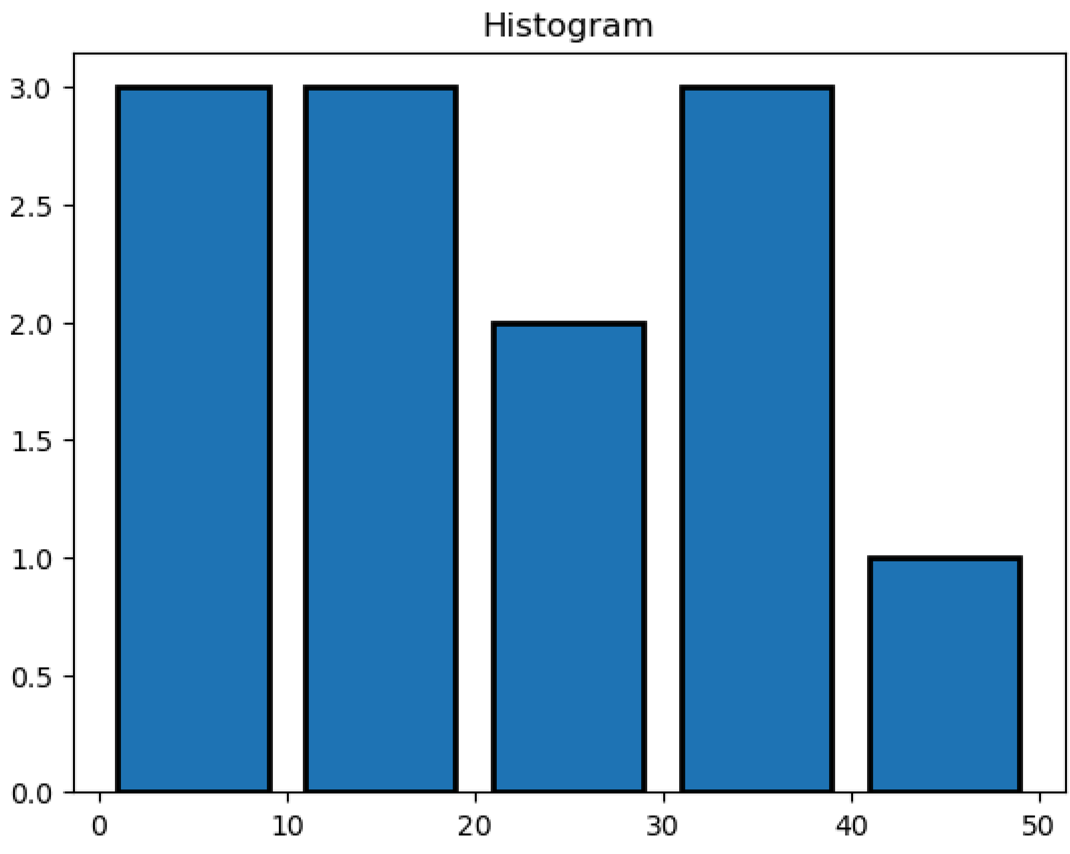





```
No artists with labels found to put in legend.  Note that artists whose label start with an un
derscore are ignored when legend() is called with no argument.
```

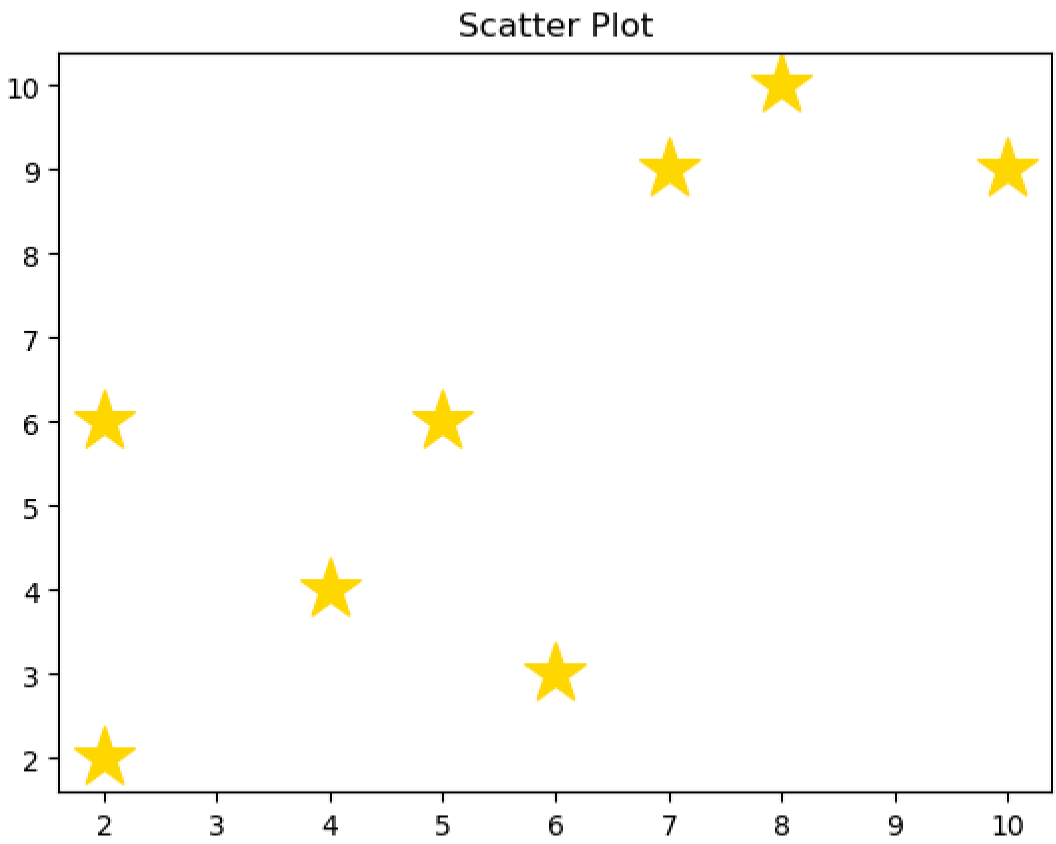


In [77]:
```
#Histogram
ages = [10,6,16,20,3,8,25,18,35,43,30,38]
bins = [0,10,20,30,40,50]

plt.hist(ages,bins,histtype='bar',rwidth=0.8,edgecolor='k',linewidth=2)
plt.title("Histogram")
plt.show()
```



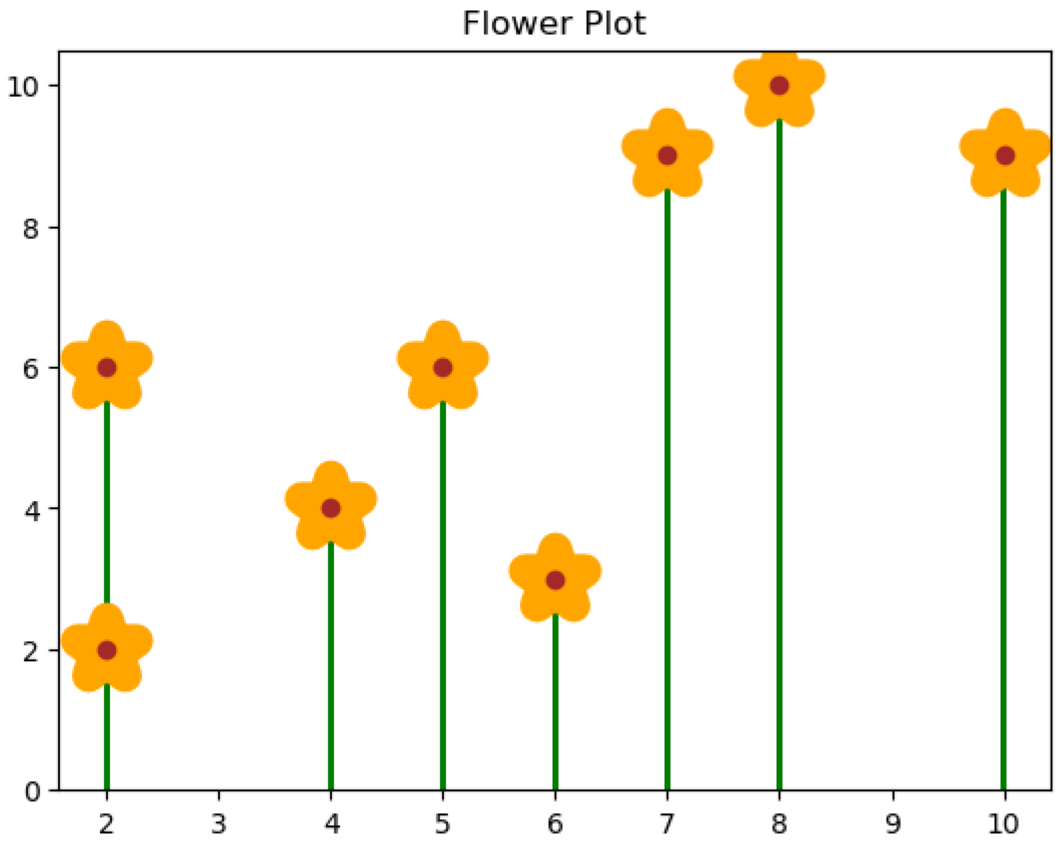
In [102…
```
#Scatter Plot
x = [2,5,4,7,2,10,6,8]
y = [2,6,4,9,6,9,3,10]
```

```
plt.scatter(x,y,s=500,color='gold',marker='*')
plt.title("Scatter Plot")
plt.show()
```



In [118…

```
x = [2,5,4,7,2,10,6,8]
y = [2,6,4,9,6,9,3,10]

plt.bar(x,y,color='g',width=0.05)
plt.scatter(x,y,s=500,color='gold',marker='*',edgecolors='orange',linewidths=12)
plt.scatter(x,y,color='brown')

plt.title("Flower Plot")
plt.show()
```



In [ ]: